Chapter Two

The Real and the Complex Number System

The real number systems

Recall:

- **Natural No: 1 , 2, 3,. . .**
- **Whole No: 0, 1, 2 , 3 . . .**
- **Integer No: . . . -3, -2, -1, 0 , 1, 2, 3, …**
- **Rational Number which is denoted by Q that is defined by** $\{\frac{p}{q}$, p, q ε Z where q≠ 0$\}$
- **Real Number** is the Union of rational and irrational numbers
- Complex number

characteristics	Natural	Whole	Z	Q	R
Closure	+ & x	+ & x	+, - & x	+, - ,÷& x	+, - ,÷ & x
commutative	+ & x	+ & x	+ & x	+ & x	+ & x
Associative	+ & x	+ & x	+ & x	+ & x	+ & x
Additive Identity	no	a=0	a=0	a=0	a=0
Additive Inverse	n0	no	All= -a	All= -a	All= -a
Multiplicative Identity	a=1	a=1	a=1	a=1	a=1
Multiplicative Inverse	no	no	no	All= 1/a	All= 1/a

The order property

1. Trichotomy:

If *x* and *y* are numbers, exactly one of the following holds:

$$x < y \; or \; x > y \; or \; x = y$$

2. Transitivity: $x < y \; and \; y < z \; then, \; x < z$

3. **Addition**:

$x < y, \quad x + z < y + z$

4. **Multiplication**:

When *z* is positive, $x < y$, $xz < yz$,

When *z* is negative, $x < y$, $xz > yz$

Greatest Common Factor (GCF)

Definition : The greatest common factor (GCF) of two numbers *a* & *b* is denoted by GCF (*a*, *b*) and is the **greatest number which is a factor** of each of the given number.

Example: Consider the two numbers 24 and 60.

F_{24} = { 1,2,3,4,6,8,12,24}

F_{60} = { 1,2,3,4,6,8,10,12,15, 20, 30, 60}

$F_{24} \cap F_{60}$ = { 1,2,3,4,6,8, 12} from this the greatest is 12

Therefore, GCF (24, 60) = 12.

Hence an alternative method (**Prime factorization method**) is provided as below:

Step 1: Find the prime factorization of each of the natural numbers

Step 2: Form the GCF of the given numbers as the product of every factor that appears in each of the prime factorization but take the least number of times it appears.

Example: Consider the two numbers 24 and 60.

$24 = 2^3 \times 3$

$60 = 2^2 \times 3 \times 5$

The factors that appear in both cases are 2 and 3, but take the numbers with the least number of times.

Therefore *GCF* (24, 60) = $2^2 \times 3 = 12$

Least Common Multiple (LCM)

Definition : The least common multiple (LCM) of two numbers *a* & *b* is denoted by

LCM (*a*, *b*) and is the **least number which is a multiple** of each of the given number.

Example : Consider the two numbers 18 and 24.

M18= { 18, 36, 54 , 72 … }

M24 = { 24, 48, 72…}

M18 $\cap$ M24 = {72 , 144…}

Therefore, LCM (18, 24) = 72

The method of finding the LCM of two or more numbers is usually lengthy and time consuming. Hence an alternative method **(Prime factorization method)** is provided as below:

Step 1: Find the prime factorization of each of the natural numbers

Step 2: Form the LCM of the given numbers as the product of every factor that appears in any of the prime factorization but take the highest number of times it appears.

Example : Consider the two numbers 18 and 24

$18 = 3^2 X\ 2$

$24 = 2^3 X\ 3$

The factors that appear in any case are 2 and 3, but take the numbers with the highest number of times.

$LCM\ (18, 24) = 2^3 X\ 3^2 = 72$

Example Find LCM of 15 and 24

- $15 = 2^0 \times 3^1 \times 5^1$
- $24 = 2^3 \times 3^1 \times 5^0$

$$\text{LCM} = 8 \times 3 \times 5$$

$$\text{LCM} = 24 \times 5 = 120$$

Principle of Mathematical Induction

Mathematical induction is one of the most important techniques used to prove mathematical problems.

For a given statement involving a natural number n, if

- The assertion is true for n = 1 (usually).
- It is true for n = k+1, whenever it is true for n = k ($k \geq 1$),

then the assertion is true for every natural number n.

The method is used to prove different propositions involving positive integers using **three** steps:

Step1: Prove that T_k(usually T_1) holds true.

Step 2: Assume that T_k for k = n is true.

Step 3: Show that T_k is true for k = n+1.

Example: **Show 1+ 3+ 5 . . . + (2n -1) = n^2**

Step 1: For n= 1, 1 = 1^2

Step 2: *Assume that it is true for* n = k

1+ 3+ 5 . . . + (2k -1) = k^2

Step 3: We should show for n= k+1

1+ 3+ 5 . . . + (2k -1) +(2(k+1) -1) = $(k+1)^2$

1+ 3+ 5 . . . + (2k -1) +2k+1 = $(k+1)^2$

$\underline{1+ 3+ 5 . . . + (2k -1)}$ +(2k+1) = $k^2 + 2k+1$

k^2

= $(k+1)^2$ Which is required.

Therefore It is true for any natural number n

Activity *: Show that 1+ 2 + 3 . . . + (n) =* $\frac{n(n+1)}{2}$

*Step*1. *For* $n = 1, \quad 1 = \dfrac{1(1+1)}{2}$ *which is true.*

*Step*2. *Assume that it is true for* $n = k$

i.e. $1 + 2 + 3 + \cdots + (k) = \dfrac{k(k+1)}{2}.$

*Step*3. *We should show that it is true for* $n = k + 1$

$$\textit{Claim}: 1 + 2 + 3 + \cdots + (k) + (k+1) = \frac{(k+1)(k+2)}{2}.$$

$$\textit{Now} \quad \underline{1 + 2 + 3 + \cdots + (k)} + (k+1) = \frac{k(k+1)}{2} + (k+1)$$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

$$= \underline{\underline{\frac{(k+1)(k+2)}{2}}} \quad \textit{which is the required result.}$$

$\therefore$ *It is true for any natural number* n.

Decimal numbers

1. Terminating decimals

Consider any terminating decimal number d.

Suppose d terminates n digits after the decimal point.

Then, d can be converted to its **fraction** form as below:

$$d = dX1 = d \times \frac{1}{1} = d \times \frac{(10)^n}{(10)^n}$$

Example : Convert the terminating decimal 3.47 to fraction form.

Solution: $$3.47 = 3.47 * \frac{10^2}{10^2} = \frac{347}{100}$$

2. Non-terminating periodic decimals

Consider any **non-terminating periodic** decimal number d.
Suppose d has **k non-terminating** digits and **p terminating** digits after the decimal point.
Then, d can be converted to its fraction form as below:

$$\mathbf{d = d * 1 = d * \frac{1}{1} = d * \frac{10^{k+p} - 10^{p}}{10^{k+p} - 10^{p}}}$$

Example :
Convert the non-terminating periodic decimal 42.5383838 to fraction form.
K=2 p=1

$$42.5383838 * \frac{10^{1+2} - 10^{1}}{10^{1+2} - 10^{1}} = 42.5383838 * \frac{10^{3} - 10^{1}}{10^{3} - 10^{1}} \frac{42538.3838 - 425.383838}{1000 - 10}$$

$$= \frac{42113}{990}$$

3. Non-terminating and non-periodic decimals

Some decimal numbers are neither terminating nor non-terminating periodic. Such types of numbers are called irrational numbers.

Example 2.40: Show that $\sqrt{2}$ is an irrational number.

Proof:

$$\text{Suppose } \sqrt{2} \text{ is a rational number}$$

$$\Rightarrow \sqrt{2} = \frac{a}{b}, \text{ where } GCF(a,b) = 1$$

$$\Rightarrow 2 = \frac{a^2}{b^2}$$

$$\Rightarrow a^2 = 2b^2 \ldots\ldots\ldots (*)$$

$$\Rightarrow a^2 \text{ is even}$$

$$\Rightarrow a \text{ is even}$$

$$\Rightarrow a = 2n \ldots\ldots\ldots\ldots (**)$$

Putting this in (*) we get :

$\Rightarrow 4n^2 = 2b^2$

$\Rightarrow b^2 = 2n^2$

$\Rightarrow b^2$ *is even*

$\Rightarrow b$ *is even*

$\Rightarrow b = 2m$.....(***)

From (**) and (***) we get a contradiction that GCF $(a, b) = 1$ which implies that $\sqrt{2}$ is not a rational number.

Therefore, $\sqrt{2}$ is an irrational number.

Intervals

Let *a* and *b* be two real numbers such that $a < b$, then the intervals which are subsets of $\boldsymbol{R}$ with end points a and b are denoted and defined as below:

i) (a, b) $= \{ x : a < x < b \}$ open interval from *a* to *b*.

ii) $[a , b] = \{ x : a \leq x \leq b \}$ closed interval from *a* to *b*.

iii) $(a, b] = \{ x : a < x \leq b \}$ open-closed interval from *a* to *b*.

iv) $[a, b) = \{ x : a \leq x < b \}$ closed-open interval from *a* to *b*.

Definition 2.11

Let A be non−empty and $A \subseteq \Re$.

i. A point $a \in R$ is said to be an upper bound of A iff $x \leq a$ for all $x \in A$.

ii. An upper bound of A is said to be least upper bound (lub) iff it is the least of all upper bounds.

iii. A point $a \in R$ is said to be lower bound of A iff $x \geq a$ for all $x \in A$.

ii. A lower bound of A is said to be greatest lower bound (glb) iff it is the greatest of all lower bounds.

Consider the set $A = \{-2, 0, 3, 5\}$. Find all lower bounds, the greatest lower bound (infimum), all upper bounds, and the least upper bound (supremum).

Solution:

- **Lower Bounds:** A number l is a lower bound if $l \leq x$ for all $x \in A$. The smallest element in A is -2. So, all numbers ≤ -2 are lower bounds.
- **Greatest Lower Bound (Infimum):** The largest of these lower bounds is -2. So, $\inf(A) = -2$. Since -2 is also in the set, it is also the **minimum**.
- **Upper Bounds:** A number u is an upper bound if $u \geq x$ for all $x \in A$. The largest element in A is 5. So, all numbers ≥ 5 are upper bounds.
- **Least Upper Bound (Supremum):** The smallest of these upper bounds is 5. So, $\sup(A) = 5$. Since 5 is also in the set, it is also the **maximum**.

Answer:

Lower bounds: $(-\infty, -2]$, $\inf(A) = -2$, Upper bounds: $[5, \infty)$, $\sup(A) = 5$.

Consider the set $B = \{x \in \mathbb{R} : 1 < x < 4\}$. Find its lower bounds, infimum, upper bounds, and supremum. Does this set have a minimum or maximum element?

Solution:

- **Lower Bounds:** Any number ≤ 1 is a lower bound. For example, 1, 0, -100. (Note: 1.5 is *not* a lower bound because, for instance, 1.1 is in B and 1.1 < 1.5).
- **Greatest Lower Bound (Infimum):** The largest number that is still a lower bound is 1. So, $\inf(B) = 1$.
- **Upper Bounds:** Any number ≥ 4 is an upper bound.
- **Least Upper Bound (Supremum):** The smallest number that is still an upper bound is 4. So, $\sup(B) = 4$.
- **Minimum/Maximum:** The set B contains numbers arbitrarily close to 1 and 4, but it does **not** contain 1 or 4 themselves. Therefore, there is **no minimum** and **no maximum** element.

Consider the set $D = \mathbb{Z}$, the set of all integers $\{\dots, -2, -1, 0, 1, 2, \dots\}$. Discuss its bounds.

Solution:

- **Upper Bounds:** Is there a number that is greater than or equal to *every* integer? No. For any candidate u, the integer $\lceil u \rceil + 1$ is larger than u. Therefore, there are **no upper bounds**.
- **Least Upper Bound (Supremum):** Since there are no upper bounds, the concept of a "least" one does not apply. We say the supremum **does not exist**.
- **Lower Bounds:** Similarly, is there a number that is less than or equal to *every* integer? No. For any candidate l, the integer $\lfloor l \rfloor - 1$ is smaller than l. Therefore, there are **no lower bounds**.
- **Greatest Lower Bound (Infimum):** Since there are no lower bounds, the infimum **does not exist**.

Consider the set $E = \{x \in \mathbb{R} : x^2 + x < 6\}$. Find its infimum and supremum.

Solution:

First, we need to understand what this set is.

1. Solve the related equation: $x^2 + x = 6$
   $x^2 + x - 6 = 0$
   $(x + 3)(x - 2) = 0$
   So, $x = -3$ or $x = 2$.
2. Since the parabola $x^2 + x - 6$ opens upwards, the region where it is less than zero (< 6) is between the roots.
3. Therefore, $E = \{x \in \mathbb{R} : -3 < x < 2\}$. This is an open interval.

Now we can analyze the bounds:

- **Lower Bounds:** All numbers ≤ -3.
- **Greatest Lower Bound (Infimum):** $\inf(E) = -3$.
- **Upper Bounds:** All numbers ≥ 2.
- **Least Upper Bound (Supremum):** $\sup(E) = 2$.

The set of complex numbers

Definition: The set of complex numbers is denoted by $\mathbb{C}$ and is described by

$$\mathbb{C} = \{z / z = x + iy,\ x, y \in \Re \text{ and } i^2 = -1\}$$

x is called the real part denoted by Re(Z)

And *y is called the imaginary part and denoted by Im(Z)*

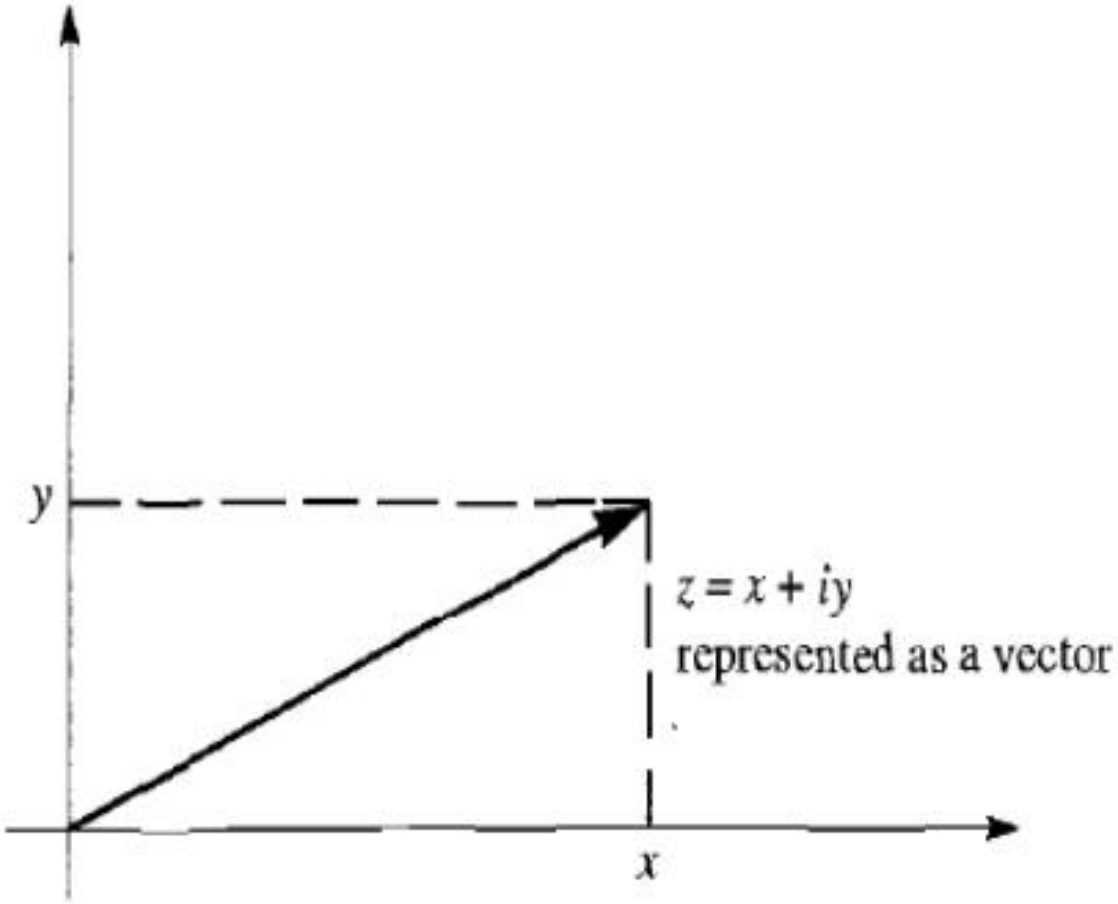
Example : *Find the real & imaginary part of the following complex numbers* :

a. Z= 3+7i

b. z = 1+i

Plotting complex numbers

Any Complex Number Z= x + yi can be draw in the complex plane as below



Equality of two complex Number

Two complex numbers Z_1 = a+bi and Z_2 = c+di *are equal iff a=c and b=d*

Example

*If z*1 = 2 + *ix and z*2 = *y* + 6*i are equal, then find the value of x* & *y*.

Operations on Complex numbers

Let Z_1 = a + bi and Z_2 = c + di be any two complex Number then

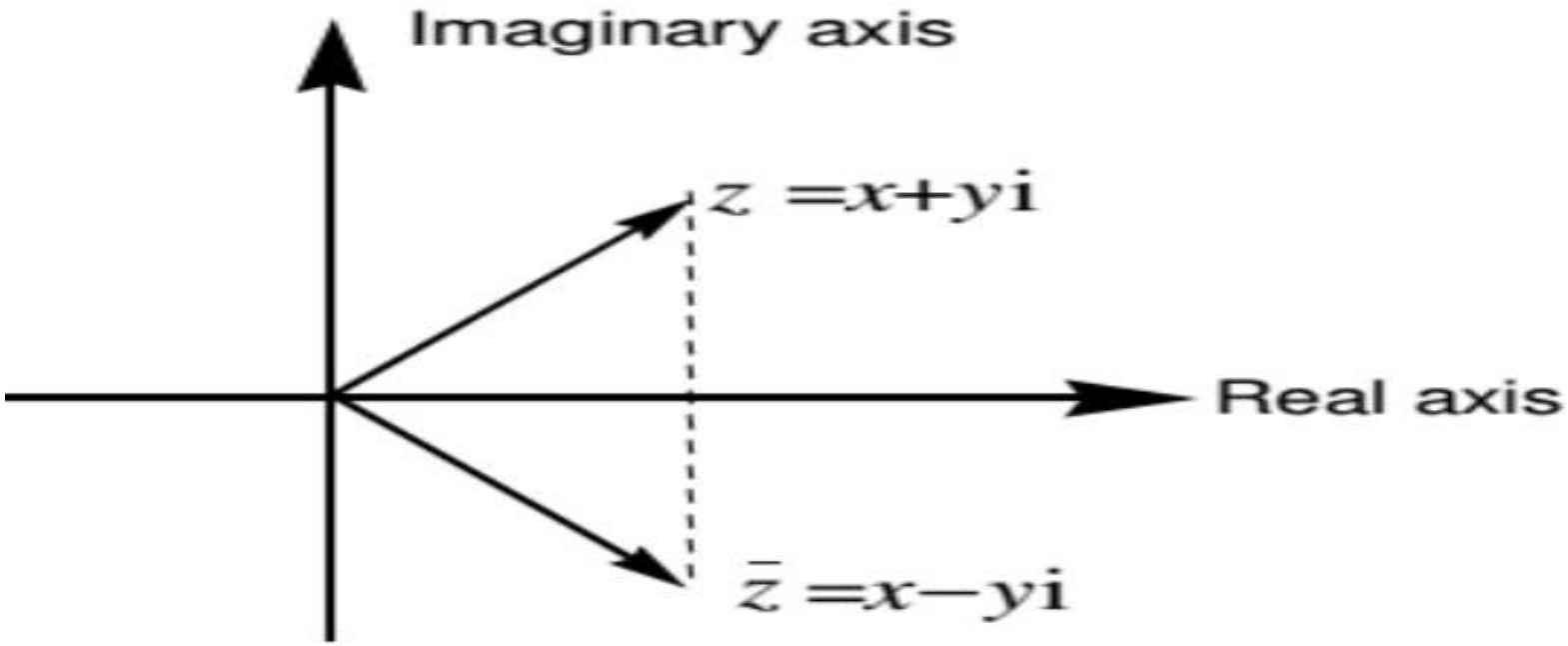
- $Z_1 + Z_2$ = (a + c) + (b+d)i
- $Z_1 - Z_2$ = (a - c) + (b- d)i
- $Z_1 * Z_2$ = (a + bi) * (c + di) = (ac- bd) + (ad + bc)I
- $\frac{Z_1}{Z_2} = \frac{a + bi}{c + di}$, $Z_2 \neq 0$

Example If Z_1 = 2 + 3i and Z_2 = 4+ i then

find a. $Z_1 + Z_2$ b. $Z_1 - Z_2$ c. $Z_1 * Z_2$ $d.$ $\frac{Z_1}{Z_2}$

Conjugate of a complex number

The conjugate of a complex number z = x+iy is denoted by $\overline{Z}$ and is defined as $\overline{Z}$ = x- iy. It can be represented by the point (x, -y) which is the reflection of the point (x, y) about the x-axis.



Properties of Conjugate

$$a.\ \bar{\bar{z}} = z$$

$$b.\ z + \bar{z} = 2x = 2\,\mathrm{Re}(z) = 2\left(\frac{z+\bar{z}}{2}\right)$$

$$c.\ z - \bar{z} = 2iy = 2i\,\mathrm{Im}(z)$$

$$d.\ \overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$$

$$e.\ \overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$$

$$f.\ \overline{z_1 . z_2} = \bar{z}_1 . \bar{z}_2$$

$$g.\ \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$$

Modulus (Norm) of a complex number

The modulus of a complex number $z = x + iy$ is a non-negative real number denoted by $|z|$ and is defined as $|z| = \sqrt{x^2 + y^2}$.

Geometrically, the number $|z|$ represents the distance between the point (x, y) and the origin.

Example : Find the modulus of the complex number z = 3 – 4i.

Properties of modulus

a. $|z| = |\bar{z}|$

b. $|z|^2 = z.\bar{z}$

c. $|z_1.z_2| = |z_1|.|z_2|$

d. $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$

e. $|z_1 + z_2| \leq |z_1| + |z_2|$ .....*triangle inequality*

f. $|z_1 - z_2| \geq |z_1| - |z_2|$

Additive and multiplicative inverses

Let $z = x+iy$ be a complex number, then

i) its additive inverse denoted by $(-z)$ *is given by:* $-z = -(x+iy) = -x-iy.$

ii) its multiplicative inverse denoted by z^{-1} *is given by:* $z^{-1} = \frac{1}{x+iy} = \frac{x}{x^2+y^2} - \frac{iy}{x^2+y^2}.$

Argument (Amplitude) of a complex number

Argument of a complex number $z = x+iy$ is the angle formed by the complex number $z = x+iy$ with the positive x-axis. The argument of a complex number $z = x+iy$ is deonted by **argz** and is given by $\arg(z) = \tan^{-1}(y/x)$.

The particular argument of z that lies in the range $-\pi < \theta \leq \pi$ is called the principal argument of z and is dented by Argz.

1. Argument of a complex number is not unique, since if θ be a value of the argument, then $2n\pi + \theta, n \in \mathbb{Z}$ is also an argument.
2. Principal argument of a complex number is unique, since principal argument always lies in the interval $(-\pi, \pi]$

Remark:

- Arg Z ϵ (-π , π]
- If 0 ≤ Arg Z ≤ , move counter clock wise direction, if not move the other direction

Example 2.50: Find the principal argument of the following complex numbers:

$a)\ z = 1 + i$ $b)\ z = -2 + 2\sqrt{3}i$ $c)\ z = -\sqrt{3} - i$

Sol : $a)\ z = 1 + i$

$$Argz = \tan^{-1}(1/1) = \tan^{-1}(1) = \frac{\pi}{4}$$

$b)\ z = -2 + 2\sqrt{3}\,i$

$$Argz = \tan^{-1}(2\sqrt{3}/-2) = \tan^{-1}(-\sqrt{3}) = \frac{2\pi}{3}$$

$c)\ z = -\sqrt{3} - i$

$$Argz = \tan^{-1}(-1/-\sqrt{3}) = \tan^{-1}(1/\sqrt{3}) = \frac{-5\pi}{6}$$

How to Find the Argument of a Complex Number
To calculate the argument of a complex number $z = a + bi$:

1. First calculate $\theta=\tan^{-1}(b/a)$.
2. If the complex number is in quadrant 1, the argument is equal to θ.
3. If the complex number is in quadrant 2, the argument is equal to $\pi-\theta$.
4. If the complex number is in quadrant 3, the argument is equal to $\theta-\pi$.
5. If the complex number is in quadrant 4, the argument is equal to $-\theta$.

Properties of Arguments

$$i)\ Arg(z_1.z_2) = Argz_1 + Argz_2 \qquad ii)\ Arg(\frac{z_1}{z_2}) = Argz_1 - \arg z_2$$

Example 2.51: Find the principal argument of $a)\ (1+i)(-1-i)$ $b)\ (\frac{-2+2i}{1-i})$

Solution:

$$a)\ Arg\,(1+i)(-1-i) = Arg\,(1+i) + Arg(-1-i) = \pi/4 + (-3\pi/4) = \underline{\underline{-\pi/2}}$$

$$b)\ Arg\,(\frac{-2+2i}{1-i}) = Arg\,(-2+2i) - Arg\,(1-i) = 3\pi/4 - (-\pi/4) = 4\pi/4 = \underline{\underline{\pi}}$$

2.2.7 Polar form of a complex number

Definition 2.16

Let r and θ be polar coordinates of the point (x, y) of the complex number z = x+iy. Since x = $r\cos\theta$ and y = $r\sin\theta$, then the complex number can be written as : $z = r(\cos\theta + i\sin\theta)$ which is called polar form, where r is modulus of z and θ is principal argument of z.

DeMoivre's Formula

Computing high powers of complex numbers
Finding roots of complex numbers
Deriving trigonometric identities
Solving equations involving complex numbers

recall the product: $z_1.z_2 = r_1.r_2[\cos(\theta_1+\theta_2)+i\sin(\theta_1+\theta_2)]$.

Similarly, we get $z_1.z_2.\cdots.z_n = r_1.r_2.\cdots.r_n[\cos(\theta_1+\theta_2+\cdots+\theta_n)+i\sin(\theta_1+\theta_2+\cdots+\theta_n)]$.

Now we can generalize that $z^n = r^n[\cos(\theta+\theta+\cdots+\theta)+i\sin(\theta+\theta+\cdots+\theta)]$

$= r^n[\cos n\theta + i\sin n\theta]$ *which is called De Moivre's formula.*

Example 2.55: Express $(2+2i)^{100}$ in polar form.

Solution: *Let* $z = 2+2i$. *Then,* $r = \sqrt{8}$, $\theta = \pi/4$ *and hence*

$$(2+2i)^{100} = z^{100}$$

$$= \sqrt{8}^{100}\left[\cos 100(\pi/4) + i \sin 100(\pi/4)\right]$$

$$= \underline{\underline{8^{50}[\cos 25\pi + i \sin 25\pi].}}$$

Example 2.56: Express $(\sqrt{3}+i)^{60}$ in polar form.

Solution: *Let* $z=\sqrt{3}+i$. *Then*, $r=2$, $\theta=\pi/6$.

$$\therefore\ (\sqrt{3}+i)^{60}=z^{60}$$

$$=2^{60}[\cos 60(\pi/6)+i\sin 60(\pi/6)]$$

$$=\underline{\underline{2^{60}[\cos 10\pi+i\sin 10\pi]}}.$$

Euler's formula

The complex number $z = r(\cos\theta + i\sin\theta)$ can be written in exponential form as: $z = re^{i\theta}$ which is called **Euler's formula**

Note: $Z^n = r^n(\cos n\theta + i\sin n\theta) = r^n e^{i(n\theta)}$

Example 2.57: Express the complex number z = 1+i using Euler's formula.

Solution: $z = 1 + i$

Now $r = \sqrt{2}$ & $\theta = \pi/4$ $\Rightarrow z = re^{i\theta} = \underline{\underline{\sqrt{2}e^{\pi i/4}}}$

Example 2.58: Express the complex number $z = 1 + \sqrt{3}\,i$ using Euler's formula.

Roots of complex numbers (Extraction of roots)

$$\text{Suppose } z_o = r_o e^{i\theta_o} \text{ is the } n^{th} \text{ root of a non-zero complex number } z = re^{i\theta}, \text{ where } n \geq 2.$$

$$\text{Then, } z_o^{\,n} = z, \text{ which implies that } r_o^{\,n} e^{in\theta_o} = re^{i\theta}$$

$$\Rightarrow r_o^{\,n} = r \;\; \& \;\; n\theta_o = \theta + 2k\pi, \quad k = 0, 1, 2, \cdots (n-1).$$

$$\Rightarrow r_o = (r)^{1/n} \;\; \& \;\; \theta_o = \frac{\theta}{n} + \frac{2k\pi}{n}$$

$$\therefore \quad z_o = \underline{\underline{(r)^{1/n}\left(e^{i\left(\frac{\theta}{n}+\frac{2k\pi}{n}\right)}\right)}} \text{ which is the } n^{th} \text{ root of } z, \text{ where } n = 2, 3, \cdots \text{ and } k = 0, 1, 2, \cdots (n-1)$$

$$\text{or we can denote it by } C_k \text{ as : } C_k = \underline{\underline{(r)^{1/n}\left(e^{i\left(\frac{\theta}{n}+\frac{2k\pi}{n}\right)}\right)}}, \quad k = 0, 1, 2, \cdots (n-1)$$

Example : Find the square roots of the complex number $z = 1+\sqrt{3}\,i$

Solution:

$z = 1+\sqrt{3}\,i$

Here $r = 2,\ \theta = \pi/3$

Hence $C_k = (r)^{1/n}(e^{i(\frac{\theta}{n}+\frac{2k\pi}{n})}),\ n = 2\ ,\ \ k = 0, 1.$

$$\Rightarrow C_k = (2)^{1/2}(e^{i(\frac{\pi/3}{2}+\frac{2k\pi}{2})})$$

$$\Rightarrow C_k = \sqrt{2}(e^{i(\pi/6+k\pi)})$$

i.) *If* $k = 0,\ C_o = \sqrt{2}(e^{i(\pi/6)}) = \sqrt{2}\,(\cos\pi/6 + i\sin\pi/6) = \sqrt{2}(\frac{\sqrt{3}}{2}+\frac{i}{2}) = \frac{\sqrt{2}}{2}(\sqrt{3}+i) = \underline{\underline{\frac{\sqrt{6}}{2}+\frac{\sqrt{3}\,i}{2}}}$

ii) *If* $k = 1,\ C_1 = \sqrt{2}(e^{i(\pi/6+\pi)}) = \sqrt{2}\,(\cos 7\pi/6 + i\sin 7\pi/6) = \sqrt{2}(\frac{-\sqrt{3}}{2}-\frac{i}{2}) = \frac{-\sqrt{2}}{2}(\sqrt{3}+i) = \underline{\underline{\frac{-\sqrt{6}}{2}-\frac{\sqrt{3}\,i}{2}}}$

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$\therefore$ *The square roots of* $1+\sqrt{3}i$ *are* $C_0 = \frac{\sqrt{6}}{2} + \frac{\sqrt{3}i}{2}$ & $C_1 = \frac{-\sqrt{6}}{2} - \frac{\sqrt{3}i}{2}$.

Example 2.61: Find the cube roots of the complex number $z = 8i$.

Solution: *We have* $z = 8i$. *Here* $r = 8,\ \theta = \pi/2,\ n = 3,\ k = 0, 1, 2.$

Hence, $C_k = (r)^{1/n}\left(e^{i\left(\frac{\theta}{n} + \frac{2k\pi}{n}\right)}\right)$

$$\Rightarrow C_k = (8)^{1/3}\left(e^{i\left(\frac{\pi/2}{3} + \frac{2k\pi}{3}\right)}\right)$$

$$\Rightarrow C_k = 2\left(e^{i\left(\pi/6 + \frac{2k\pi}{3}\right)}\right)$$

$$i)\ If\ \ k=0,\ C_o=2(e^{i(\pi/6)})=(\cos\pi/6+i\sin\pi/6)=2(\frac{\sqrt{3}}{2}+\frac{i}{2})=\underline{\underline{\sqrt{3}+i}}$$

$$ii)\ If\ \ k=1,\ C_1=2(e^{i(\pi/6+2\pi/3)})=2(\cos 5\pi/6+i\sin 5\pi/6)=2(\frac{-\sqrt{3}}{2}+\frac{i}{2})=\underline{\underline{-\sqrt{3}+i}}$$

$$iii)\ If\ \ k=2,\ C_2=2(e^{i(\pi/6+4\pi/3)})=2(\cos 3\pi/2+i\sin 3\pi/2)=2(0+-i)=\underline{\underline{-2i}}$$

$$\therefore\ The\ cube\ roots\ of\ \ 8i\ \ are\ \ C_o=\sqrt{3}+i\ \ ,\ \ C_1=-\sqrt{3}+i\ \ \&\ C_2=-2i.$$